Exercise 1.2

9QADE9

1. Fill in the blanks

(i) The HCF of 45 and 75 is _____.

(ii) The HCF of two successive even numbers is _____.

(iii) If the LCM of 3 and 9 is 9, then their HCF is _____.

(iv) The LCM of 26, 39 and 52 is_____.

(v) The least number that should be added to 57 so that the sum is exactly divisible by 2,3, 4 and 5 is _____.

2. Say True or False

(i) The numbers 57 and 69 are co-primes.

(ii) The HCF of 17 and 18 is 1.

(iii) The LCM of two successive numbers is the product of the numbers.

(iv) The LCM of two co-primes is the sum of the numbers.

(v) The HCF of two numbers is always a factor of their LCM .

3. Find the HCF of each set of numbers using prime factorisation method.

(i) 18,24 (ii) 51,85 (iii) 61,76 (iv) 84,120 (v) 27,45,81 (vi) 45,55,95

4. Find the LCM of each set of numbers using prime factorisation method.

(i) 6,9 (ii) 8,12 (iii) 10,15 (iv) 14,42 (v) 30,40,60 (vi) 15,25,75

5. Find the HCF and the LCM of the numbers 154, 198 and 286.

6. What is the greatest possible volume of a vessel that can be used to measure exactly the volume of milk in cans (in full capacity) of 80 litres, 100 litres and 120 litres?

7. The traffic lights at three different road junctions change after every 40 seconds, 60 seconds and 72 seconds respectively. If they changed simultaneously together at 8 a.m at the junctions, at what time will they simultaneously change together again?

8. The LCM of two numbers is 210 and their HCF is 14. How many such pairs are possible?

9. The LCM of two numbers is 6 times their HCF. If the HCF is 12 and one of the numbers is 36, then find the other number.

Objective Type Questions

10. Which of the following pairs is co-prime?

a) 51, 63 b) 52, 91 c) 71, 81 d) 81, 99

11. The greatest 4 digit number which is exactly divisible by 8, 9 and 12 is

a) 9999 b) 9996 c) 9696 d) 9936

12. The HCF of two numbers is 2 and their LCM is 154. If the difference between the numbers is 8, then the sum is

a) 26 b) 36 c) 46 d) 56

13. Which of the following cannot be the HCF of two numbers whose LCM is 120?

a) 60 b) 40 c) 80 d) 30

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